Koneru Lakshmaiah Education Foundation
(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)
Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.
Phone No: 7815926816, www.klh.edu.in

STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	I-III Semester (2025-26)
Faculty Name	Dr Sukanya Ledalla
Student Name	Billakanti Siri
Roll Number	2520030601
Branch	CSE
Course Outcome	CO3– BFS Shortest Path

Case Study Topic: Wikipedia Hyperlink Graph — Six Degrees of Separation using BFS

Work Done Summary:

In this case study, we implemented Breadth First Search (BFS) on a Wikipedia hyperlink graph to find the minimum hop-count from the article "Cricket" to all other connected articles. The graph represents Wikipedia pages as nodes and hyperlinks as edges. BFS traversal was used to explore the graph level-by-level and determine shortest paths in terms of hyperlink hops.

Implementation Details:

1.Created a graph structure using Java adjacency lists.
2.Added undirected hyperlink edges between Wikipedia articles.
3.Implemented Breadth First Search (BFS) traversal using Queue data structure.
4.Maintained visited nodes to avoid repeated traversal.
5.Displayed BFS frontier expansion level-by-level.

Result: Screenshots/output


```
====== BFS LEVEL ORDER EXPANSION ======

Current Article : Cricket
Visited : India | Hop Count : 1
Visited : Sachin | Hop Count : 1

Current Article : India
Visited : Mumbai | Hop Count : 2
Visited : Tendulkar | Hop Count : 2

Current Article : Sachin
Visited : Wankhede | Hop Count : 2

Current Article : Mumbai
Visited : Bandra | Hop Count : 3
Visited : MumbaiCity | Hop Count : 3

Current Article : Tendulkar

Current Article : Wankhede

Current Article : Bandra
```

GitHub Link:

https://github.com/siribillakanti/DSA2_ASSIGNMENTS/blob/main/assignment_CO3.java